

solucion

a) tenemos que

$$\vec{F} = e\vec{E} + \frac{e}{c} \vec{v} \times \vec{B}$$

donde $E=0$ y $\vec{v} \perp \vec{B}$

$$\rightarrow |\vec{v} \times \vec{B}| = vB$$

por lo que

$$F_B = \frac{e}{c} vB$$

como solo hay fuerza magnetica no produce trabajo
no varía la velocidad solo cambia su dirección
por lo que es un movimiento circular

Para un movimiento circular

$$F_c = \frac{mv^2}{r}$$

como la fuerza magnetica es la responsable
de la curvatura

$$\frac{mv^2}{r} = \frac{e v B}{c}$$

despejamos

$$r = \frac{mv c}{e B}$$

b)

tenemos $B = B \hat{e}_z$ $E = E_y \hat{e}_y + E_z \hat{e}_z$

$$\vec{F} = e(E_y \hat{e}_y + E_z \hat{e}_z) + \frac{e}{c} \vec{v} \times B \hat{e}_z$$

$$m \dot{\vec{z}} = e E_z \hat{e}_z + \frac{e}{c} (\vec{v} \times B) \hat{e}_z$$

Le pedimos q calcule que nos ayude con este
producto cruz

$$\vec{v} \times \vec{B} = 0$$

$$\vec{v} \times \vec{B} = \begin{vmatrix} \hat{e}_x & \hat{e}_y & \hat{e}_z \\ \dot{x} & \dot{y} & \dot{z} \\ 0 & 0 & B \end{vmatrix} = (B \dot{y} - B \dot{x}, 0)$$

$(\vec{v} \times \vec{B})_z = 0$

Norma

$$m \ddot{z} = e E z \rightarrow \ddot{z} = \frac{e E}{m} z$$

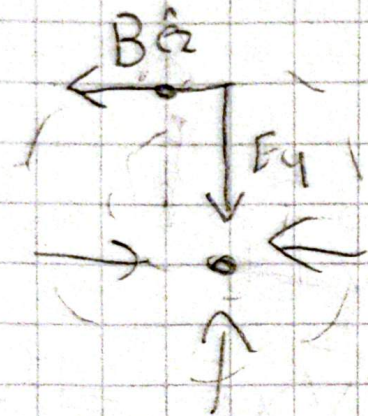
ahora integramos

$$\dot{z} = \frac{e E z}{m} + C_1, C_1 = \dot{z}_0$$

$$z(t) = z_0 + \frac{e E z}{2m} t^2 + C_2 \quad C_2 = z_0$$

$$z(t) = z_0 + \dot{z}_0 t + \frac{e E z}{2m} t^2$$

a)



$$\vec{B} = B \hat{x}$$

$$\vec{E} = E_y \hat{y} + E_z \hat{z}$$

$$m \vec{\dot{v}} = e \vec{E} + \frac{e}{c} \vec{v} \times \vec{B}$$

$$\vec{v} \times \vec{B} = B \dot{y} \hat{x} - B \dot{x} \hat{y}$$

$$m \ddot{x} = \frac{e B}{c} \dot{y}$$

$$m \ddot{y} = e E_y - \frac{e B}{c} \dot{x}$$

$$\ddot{x} = \omega_c \dot{y} \quad \ddot{y} = \frac{e E_y}{m} - \omega_c \dot{x}$$

re pethos q ta Al qur nos qyude con ta E D. 12

$$u = \dot{x}$$

$$\ddot{u} = \ddot{x} = w_c \dot{y}$$

$$\ddot{u} = w_c \dot{y} = w_c \left(\frac{c E y}{m} - w_c u \right)$$

$$\ddot{u} + w_c^2 u = w_c \frac{c E y}{m}$$

$$\frac{c E y}{m} = w_c \frac{c E y}{B}$$

$$\ddot{u} + w_c^2 u = w_c^2 \left(\frac{c E y}{B} \right)$$

$$\dot{x} = \frac{c E y}{B} + A \cos(w_c t) + C \sin(w_c t)$$

$$\dot{x}(0) = \dot{x}_0 \quad t=0$$

$$\dot{x}_0 = \frac{c E y}{B} + A \rightarrow A = \dot{x}_0 - \frac{c E y}{B}$$

$$\ddot{x} = w_c \dot{y}$$

$$\ddot{x} = -w_c \left(\dot{x}_0 - \frac{c E y}{B} \right) \sin(w_c t) + w_c C \cos(w_c t)$$

-y

$$\dot{y}(0) = C$$

$$\dot{x}(t) = \frac{c E y}{B} + \left(\dot{x}_0 - \frac{c E y}{B} \right) \cos(w_c t) + \dot{y}_0 \sin(w_c t)$$

$$\dot{y}(t) = \dot{y}_0 \cos(w_c t) - \left(\dot{x}_0 - \frac{c E y}{B} \right) \sin(w_c t)$$

quando tememos a chatopê para nos
e a gente q' interesse

$$x(t) = \int \dot{x}(t) dt$$

$$x(t) = x_0 + \frac{c E y}{B} t + \frac{\dot{x}_0 - c \frac{E y}{B}}{\omega c} \sin(\omega c t) + \frac{\dot{y}_0}{\omega c} [1 - \cos(\omega c t)]$$

$$y(t) = \int \dot{y}(t) dt$$

$$y(t) = y_0 + \frac{\dot{y}_0}{\omega c} \sin(\omega c t) + \frac{\dot{x}_0 - c \frac{E y}{B}}{\omega c} [\cos(\omega c t) - 1]$$

valor medio

$$\langle \dot{x} \rangle = \frac{1}{T} \int_0^T \dot{x}(t) dt$$

$$\langle \dot{x} \rangle = \frac{c E y}{B} + \frac{1}{T} \left(\dot{x}_0 - c \frac{E y}{B} \right) \int_0^T \cos(\omega c t) dt + \frac{\dot{y}_0}{T} \int_0^T \sin(\omega c t) dt$$

$$\langle \dot{x} \rangle = \frac{c E y}{B}$$

$$\langle \dot{y} \rangle = \frac{1}{T} \int_0^T \dot{y}(t) dt$$

$$\langle \dot{y} \rangle = 0$$

como $\int_0^T \cos(\omega c t) dt = 0$
 $\int_0^T \sin(\omega c t) dt = 0$